Arkady Miasnikov

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Software Engineer experienced in leading development projects from inception to full implementation

Backend and distributed-systems engineer with hands-on ownership of trading, security, and data-intensive platforms. Strong track record building performance-critical services in Go/Python, optimizing hot paths, and shipping resilient production systems. Experienced in cloud-native operations, observability, and cross-functional delivery from design through rollout. Tools & Technologies: Linux, Go, Python, C, AWS, Kubernetes, SQL, ClickHouse, Redis, Kafka.

Professional Experience

2023-present: Software Engineer - qSpark (HFT trading platform)

- Maintained and extended a mid-size Django/FastAPI backend serving US regulatory requirements, production incident response, and on-call operations.
- Rebuilt the end-to-end (e2e) test harness from scratch - now the primary validation layer for AI-generated trading components. Designed for maximum parallel execution of real trading flows against a live Kubernetes stack.
- Conceived and implemented an Airflow-based log anomaly detector (Markov fuzzing + Jaccard clustering) that surfaces noisy warning/error pattern shifts across production logs - praised internally and recognized as a significant observability contribution.
- Added Python linter, JS/TS linter, and Django DB migration check jobs to the GitLab CI pipeline - all self-initiated and self-delivered.
- Contributed to automatic liquidation flow (EMS) and Django version migrations across multiple services.
- Drove internal adoption of AI coding tools - established KPI tracking of AI contribution to MRs (later adopted org-wide as a project metric).

Technologies & Programming Languages

Python, Django, FastAPI, Kubernetes, Kafka, Redis, Elasticsearch, Airflow, GitLab CI, PostgreSQL

2023-present: Software Engineer - Endotech

- Designed and implemented custom momentum indicators and market-condition filters used in production trading strategies.
- Ported and optimized multiple indicators from PineScript to Go to improve runtime efficiency and maintainability.
- Delivered reusable signal components that became core building blocks in the company's algorithmic trading stack.

Technologies & Programming Languages

Go, Python, ClickHouse, SQL, AI

2022-2023: Software Engineer - HiAuto

- Maintained and improved a computerized drive-thru order-taking platform in a distributed international engineering team.
- Integrated operational data into analytics workflows and resolved data quality/system issues to improve business visibility.
- Supported continuous operation of on-premises devices through proactive troubleshooting and incident resolution.

Technologies & Programming Languages

Python, BigQuery, Redis, SQL, AI, k8s, Azure, Docker

2018-2022: Software Engineer - Cyren

- Designed and developed core components of email and network security products, improving threat detection quality.
- Maintained and evolved a complex SaaS platform with integrations across multiple internal and third-party services.
- Built phishing and spam detection engines used in production security pipelines.
- Drove technical POCs including headless browser analysis, image-recognition services, fast Hamming-distance processing, and locality-sensitive hashing.
- Contributed to a security pipeline processing 50M+ emails/day (~250 TB/day).

Technologies & Programming Languages

Go, C++, Python, ElasticSearch, Redis, Apache Kafka, Prometheus, Grafana, Kibana, Megalog, Jaeger, SQL, k8s, AWS, Docker, AI

2016-2018: Software Engineer - Secdo

- Defined and implemented Linux kernel probes with SystemTap and eBPF to monitor user-space activity with minimal overhead.
- Built kernel-to-cloud telemetry pipeline (eBPF/SystemTap → Kafka → Vertica) handling hundreds of millions of system events/second across multi-node enterprise environments.
- Implemented Windows kernel and driver-side collection components with user-space integration for efficient telemetry ingestion.
- Optimized detection data-processing workflows and developed behavioral models with the security research team.

Technologies & Programming Languages

Python, C/C++, Linux Kernel, SystemTap, eBPF, Windows kernel, Vertica, SQL

2012-2016: Software Engineer - Megabridge

- Developed HLS video monitoring/control systems, including hardware verification and low-level embedded firmware.
- Designed device drivers and major firmware components for the BDSL product line, including Java-based management interfaces.

- Led hardware bring-up for multiple platforms and customized Linux root filesystems with Yocto.
- Established automated build infrastructure and CI processes for faster, repeatable delivery.

Technologies & Programming Languages

C/C++, Python, Linux kernel, U-boot

2009-2012: Software Engineer - Texas Instruments

- Developed firmware for 802.11 ASIC programs, including pre- and post-silicon verification of WLAN devices.
- Contributed to WLAN Linux kernel development and built utilities for firmware build/debug workflows.
- Led migration from ClearCase to Git, improving engineering productivity by 30%.
- Built an internal wiki-based search system that improved discoverability of engineering knowledge.

Technologies & Programming Languages

C/C++, Python, Linux kernel, ASIC, Bare metal

Education

B.A. in Business Administration - Netanya Academic College (2007-2009)

Software Practical Engineer - Tel Aviv University School of Practical Engineering

Physics Department - St. Petersburg State University (1987-1989)